

Phase 2 Practice Sheet

Part 1: Arrays (20 Questions)

Easy Level (Q1 - Q7)

These questions focus on understanding arrays, accessing values, basic methods, and simple iteration.

Question 1 — Find Expensive Products

```
let prices = [100, 250, 500, 150, 700];
```

Create a new array containing only prices greater than ₹300.

Scenario

You're building an e-commerce website.

Customers want to see premium products only.

What is this question asking?

Go through every price and keep only those greater than 300.

Expected Output:

```
[500, 700]
```

Concepts Tested

- Arrays
 - filter()
-

Question 2 — Last Student in Class

```
let students= ["Aman", "Ritik", "Priya", "Rahul"];
```

Print the last student's name.

Scenario

The teacher wants to know who is sitting at the end of the attendance list.

What is this question asking?

Access the last element without manually writing the index.

Concepts Tested

- Arrays
- length property

Question 3 — Add New Product

```
let products= ["Laptop", "Mouse", "Keyboard"];
```

Add `"Monitor"` to the end of the array.

Scenario

A new product arrives in inventory.

What is this question asking?

Insert an item at the end of an array.

Concepts Tested

- push()

Question 4 — Remove Last Notification

```
let notifications= [  
  "Order Placed",  
  "Order Shipped",  
  "Order Delivered"  
];
```

Remove the latest notification.

Scenario

Notifications disappear after being viewed.

What is this question asking?

Remove the last element.

Concepts Tested

- `pop()`

Question 5 — Check User Exists

```
let users= ["Aman", "Ritik", "Priya"];
```

Check if `"Ritik"` exists.

Scenario

User tries to login.

What is this question asking?

Search for a value inside an array.

Concepts Tested

- `includes()`

Question 6 — Convert Marks to Percentage

```
let marks= [80,90,70];
```

Create a new array where each mark is followed by %.

Expected:

```
["80%", "90%", "70%"]
```

What is this question asking?

Transform every value into something else.

Concepts Tested

- `map()`

Question 7 — Count Products

```
let cart= [  
  "Mouse",  
  "Keyboard",  
  "Monitor",  
  "Laptop"  
];
```

Print total number of products.

Scenario

Shopping cart badge count.

Concepts Tested

- `length`

Moderate Level (Q8 - Q14)

These questions require combining multiple concepts.

Question 8 — Student Average

```
let marks= [80,90,70,85,95];
```

Find average marks.

Scenario

Teacher wants class average.

What is this question asking?

1. Calculate total.
2. Divide by total students.

Concepts Tested

- `reduce()`
 - `length`
-

Question 9 — Even Numbers Finder

```
let numbers= [1,2,3,4,5,6,7,8];
```

Return all even numbers.

Scenario

Filter numbers for a report.

What is this question asking?

Keep only values divisible by 2.

Concepts Tested

- `filter()`
-

Question 10 — Product Search

```
let products= [
  "Laptop",
  "Mouse",
  "Keyboard",
  "Monitor"
];
```

Find the index of `"Keyboard"`.

Scenario

You need the product position.

Concepts Tested

- `indexOf()`
-

Question 11 — Total Revenue

```
let sales= [500,700,1000,300];
```

Calculate total revenue.

Scenario

Store owner wants today's earnings.

What is this question asking?

Add all numbers together.

Concepts Tested

- `reduce()`
-

Question 12 — Uppercase Usernames

```
let users= ["ritik","aman","priya"];
```

Convert every username into uppercase.

Scenario

Preparing usernames for display.

Concepts Tested

- `map()`
-

Question 13 — Find First Adult

```
let ages= [12,15,17,19,22];
```

Find the first age greater than or equal to 18.

Scenario

Age verification system.

Concepts Tested

- `find()`
-

Question 14 — Positive Number Check

```
let nums= [5,8,10,3];
```

Check if every number is positive.

Scenario

Validate data before processing.

Concepts Tested

- every()

Hard Level (Q15 - Q20)

These require logic building and problem-solving.

Question 15 — Most Frequent Number

```
let numbers= [1,2,3,2,4,2,5,1,1,1];
```

Find the number that appears the most.

Scenario

Website wants to know the most searched category.

What is this question asking?

Count occurrences and find the highest.

Concepts Tested

- Arrays
- Loops
- Counting logic

Question 16 — Second Largest Number

```
let nums= [10,50,20,80,40];
```

Find the second largest number.

Scenario

Leaderboard runner-up.

What is this question asking?

Largest is:

80

Find:

50

Concepts Tested

- sort()
- Logic

Question 17 — Remove Duplicates

```
let ids= [1,2,2,3,4,4,5,5];
```

Create a new array with unique values.

Expected:

[1,2,3,4,5]

Scenario

Duplicate user IDs should not exist.

Concepts Tested

- Arrays
 - Problem Solving
-

Question 18 — Longest Word

```
let words= [  
  "JavaScript",  
  "HTML",  
  "CSS",  
  "Programming"  
];
```

Find the longest word.

Scenario

Text analytics system.

What is this question asking?

Compare lengths of all strings.

Concepts Tested

- Loops
 - String length
-

Question 19 — Rotate Array Right

```
let nums= [1,2,3,4,5];
```

Output:

```
[5, 1, 2, 3, 4]
```

Scenario

Carousel slider rotates items.

What is this question asking?

Move last element to the beginning.

Concepts Tested

- `pop()`
- `unshift()`

Question 20 — Best Selling Product

```
let sales= [  
  "Mouse",  
  "Keyboard",  
  "Mouse",  
  "Laptop",  
  "Mouse",  
  "Keyboard"  
];
```

Find the product sold the most.

Scenario

E-commerce analytics dashboard.

What is this question asking?

Count how many times each product appears and find the highest.

Expected:

"Mouse"

Concepts Tested

- Arrays
- Counting frequency
- Objects for tracking counts
- Problem Solving

Part 2: Objects (20 Questions)

Easy Level (Q1 - Q7)

These questions focus on creating, accessing, updating, and deleting object properties.

Question 1 — Access User Name

```
let user = {  
  name: "Ritik",  
  age: 21,  
  city: "Bhopal"  
};
```

Task

Print the user's name.

Scenario

A profile page needs to display the logged-in user's name.

What is this question asking?

The data is already stored in an object.

Your task is to access one specific property.

Expected Output

```
Ritik
```

Concepts Tested

- Objects
- Dot Notation

Question 2 — Update User Age

```
let user = {  
  name: "Ritik",  
  age: 21  
};
```

Task

Update age to 22.

Scenario

Today is the user's birthday.

What is this question asking?

Update an existing property inside an object.

Expected Output

```
{  
  name: "Ritik",  
  age: 22  
}
```

Concepts Tested

- Object Updates

Question 3 — Add Country

```
let user= {  
  name: "Ritik",  
  city: "Bhopal"  
};
```

Task

Add a property:

```
country: "India"
```

Scenario

The registration form now requires country information.

What is this question asking?

Add a completely new property.

Concepts Tested

- Adding Properties

Question 4 — Delete Property

```
let user= {  
  name:"Ritik",  
  age:21,  
  password:"12345"  
};
```

Task

Delete the password property.

Scenario

Sensitive information should not be sent to the frontend.

What is this question asking?

Remove a property from an object.

Concepts Tested

- delete keyword

Question 5 — Check Property Exists

```
let product= {  
  name:"Laptop",  
  price:60000  
};
```

Task

Check if the object contains a property called `"price"`.

Scenario

Before displaying the product price, you want to make sure it exists.

Concepts Tested

- Object Keys
 - Property Checking
-

Question 6 — Print All Keys

```
let car= {  
  brand: "BMW",  
  model: "X5",  
  year: 2025  
};
```

Task

Print all property names.

Expected Output

```
[ "brand", "model", "year" ]
```

Scenario

Admin dashboard wants to know available fields.

Concepts Tested

- Object.keys()
-

Question 7 — Print All Values

```
let car= {  
  brand: "BMW",  
  model: "X5",  
  year: 2025  
};
```


Task

Print all values.

Expected Output

```
[ "BMW", "X5", 2025 ]
```

Concepts Tested

- Object.values()

Moderate Level (Q8 - Q14)

These questions combine multiple object concepts.

Question 8 — Print User Information Dynamically

```
let user= {  
  name:"Ritik",  
  age:21,  
  city:"Bhopal"  
};
```

Task

Print:

```
name :Ritik  
age :21  
city :Bhopal
```

using a loop.

Scenario

A profile card should automatically display all information.

What is this question asking?

Instead of manually writing each property, loop through the object.

Concepts Tested

- `Object.entries()`
- `for...of`

Question 9 — Employee Salary Increase

```
let employee= {  
  name: "Aman",  
  salary: 50000  
};
```

Task

Increase salary by 10%.

Scenario

Company announces annual appraisal.

What is this question asking?

Read a property, calculate a new value, update the property.

Concepts Tested

- Object Updates
- Arithmetic Operations

Question 10 — Nested Object Access

```
let user= {  
  name:"Ritik",  
  address: {  
    city:"Bhopal",  
    state:"MP"  
  }  
};
```

Task

Print:

Bhopal

Scenario

Delivery app needs city information.

What is this question asking?

Access a property inside another object.

Concepts Tested

- Nested Objects

Question 11 — Object Destructuring

```
let student= {  
  name:"Priya",  
  age:20,  
  course:"BCA"  
};
```

Task

Extract:

```
name  
age
```

into separate variables.

Scenario

You only need a few properties from a large object.

Concepts Tested

- Object Destructuring

Question 12 — Rename During Destructuring

```
let student= {  
  name:"Priya",  
  age:20  
};
```

Task

Create:

```
studentName  
studentAge
```

Scenario

Variable names should be more meaningful.

Concepts Tested

- Destructuring Renaming

Question 13 — Merge User and Address

```
let user= {  
  name:"Ritik",  
  age:21  
};  
  
let address= {  
  city:"Bhopal",  
  state:"MP"  
};
```

Task

Combine both objects.

Scenario

Backend sends user and address separately.

Frontend needs one object.

Concepts Tested

- Spread Operator

Question 14 — Count Object Properties

```
let user= {  
  name:"Ritik",  
  age:21,  
  city:"Bhopal",  
  country:"India"  
};
```

Task

Find how many properties exist.

Scenario

Dynamic form validation.

What is this question asking?

Count total keys.

Concepts Tested

- `Object.keys()`
- `length`

Hard Level (Q15 - Q20)

These require real-world logic and object manipulation.

Question 15 — Highest Paid Employee

```
let
employees= {
  aman:25000,
  ritik:50000,
  priya:45000
};
```

Task

Find the employee earning the highest salary.

Scenario

Company wants Employee of the Month based on salary.

What is this question asking?

Compare every salary and return the employee name.

Expected Output

```
"ritik"
```

Concepts Tested

- Objects
- Loops
- Comparisons

Question 16 — Most Used Programming Language

```
let votes= {  
  JavaScript:25,  
  Python:30,  
  Java:15,  
  Cpp:10  
};
```

Task

Find the language with the highest votes.

Scenario

Survey results dashboard.

What is this question asking?

Find the largest value in an object.

Concepts Tested

- Object.entries()
- Comparisons

Question 17 — Reverse Key Value

```
let countries= {  
  India:"Delhi",  
  Japan:"Tokyo",  
  France:"Paris"  
};
```

Task

Convert into:

```
{  
  Delhi:"India",  
  Tokyo:"Japan",  
  Paris:"France"  
}
```

Scenario

Search by capital city instead of country.

Concepts Tested

- Objects
- Loops
- Transformation Logic

Question 18 — Student Marks Summary

```
let marks= {  
  math:90,  
  science:80,
```



```
    english:85  
  };
```

Task

Calculate total marks.

Scenario

Generate report card.

What is this question asking?

Add all values inside the object.

Concepts Tested

- `Object.values()`
- `reduce()`

Question 19 — Find Missing Property

```
let user= {  
  name:"Ritik",  
  age:21  
};
```

Task

Check whether:

```
email
```

exists.

If not, add:

```
email:"Not Provided"
```

Scenario

Incomplete profile detection.

Concepts Tested

- Property Checking
- Conditional Logic

Question 20 — Product Inventory Analyzer

```
let inventory= {  
  mouse:25,  
  keyboard:10,  
  monitor:5,  
  laptop:2  
};
```

Task

Find:

1. Total items in stock.
2. Product with highest stock.
3. Product with lowest stock.

Scenario

Inventory management dashboard.

What is this question asking?

Analyze all values and generate useful information.

Concepts Tested

- `Object.values()`
- `Object.entries()`
- Loops
- Comparisons
- Real-world Problem Solving

Part 3: Functions (20 Questions)

Easy Level (Q1 - Q7)

These questions focus on understanding function creation, parameters, arguments, and return values.

Question 1 — Greeting Function

Create a function:

```
greet(name)
```

that prints:

```
Hello Ritik
```

when called with:

```
greet("Ritik")
```

Scenario

A website wants to greet users after login.

What is this question asking?

Learn the most basic purpose of functions:

- Accept input
- Perform an action
- Produce output

Concepts Tested

- Function Declaration
 - Parameters
 - Arguments
-

Question 2 — Add Two Numbers

Create a function:

```
add(a, b)
```

that returns the sum.

Example:

```
add(10, 20)
```

Output:

```
30
```

Scenario

Calculator application.

What is this question asking?

Receive two inputs and return a result.

Concepts Tested

- Parameters
 - Return
-

Question 3 — Find Square

Create a function:

```
square(num)
```

that returns the square of a number.

Example:

```
square(5)
```

Output:

```
25
```

Scenario

Math utility app.

Concepts Tested

- Functions
 - Return Values
-

Question 4 — Check Even Number

Create a function:

```
isEven(num)
```

Return:

```
true
```

if the number is even, otherwise:

```
false
```

Scenario

Number validation system.

What is this question asking?

Return a boolean based on a condition.

Concepts Tested

- Functions
- Conditionals

Question 5 — Default Username

Create a function:

```
greet(name="Guest")
```

Example:

```
greet()
```

Output:

```
Hello Guest
```

Scenario

User visits without logging in.

Concepts Tested

- Default Parameters

Question 6 — Celsius to Fahrenheit

Create a function:

```
convertTemp(celsius)
```

Formula:

```
(celsius*9/5)+32
```

Scenario

Weather application.

Concepts Tested

- Functions
- Return Values

Question 7 — Arrow Function Practice

Convert this function:

```
function multiply(a,b) {  
  return a*b;  
}
```

into an arrow function.

Scenario

Modern JavaScript codebase.

Concepts Tested

- Arrow Functions

Moderate Level (Q8 - Q14)

These questions require combining multiple function concepts.

Question 8 — Discount Calculator

Create:

```
calculateDiscount(price)
```

Apply:

```
10%
```

discount.

Example:

```
calculateDiscount(500)
```

Output:

```
450
```

Scenario

E-commerce website.

What is this question asking?

Accept input.

Perform calculation.

Return result.

Concepts Tested

- Functions
 - Return
-

Question 9 — Largest of Three Numbers

Create:

```
findLargest(a, b, c)
```

Return the largest number.

Example:

```
findLargest(10, 50, 20)
```

Output:

```
50
```

Scenario

Ranking system.

Concepts Tested

- Functions
 - Comparisons
-

Question 10 — Reverse String

Create:

```
reverseString(str)
```

Example:

```
reverseString("hello")
```

Output:

```
"olleh"
```

Scenario

Text processing application.

What is this question asking?

Manipulate text and return the reversed version.

Concepts Tested

- Functions
- Strings
- Loops

Question 11 — Count Vowels

Create:

```
countVowels(str)
```

Example:

```
countVowels("javascript")
```

Output:

3

Scenario

Text analysis tool.

Concepts Tested

- Functions
- Loops
- Conditions

Question 12 — Generate Username

Create:

```
generateUsername (name)
```

Input:

```
"Ritik Rajput"
```

Output:

```
"ritik_rajput"
```

Scenario

Registration system.

Concepts Tested

- Functions
- String Methods

Question 13 — Dynamic Sum Function

Create:

```
sum(...numbers)
```

Example:

```
sum(1, 2, 3, 4)
```

Output:

```
10
```

Scenario

Calculator app.

What is this question asking?

Accept unlimited numbers.

Concepts Tested

- Rest Parameters
- `reduce()`

Question 14 — Login Validation

Create:

```
login(username, password)
```

Rules:

```
username === "admin"
```

```
password==="1234"
```

Return:

```
"Login Successful"
```

or

```
"Invalid Credentials"
```

Scenario

Authentication system.

Concepts Tested

- Functions
- Conditional Logic

Hard Level (Q15 - Q20)

These questions require problem-solving and function composition.

Question 15 — Factorial Function

Create:

```
factorial(n)
```

Example:

```
factorial(5)
```

Output:

120

Because:

$5 \times 4 \times 3 \times 2 \times 1$

Scenario

Math engine.

Concepts Tested

- Loops
- Functions

Question 16 — Palindrome Checker

Create:

```
isPalindrome(str)
```

Example:

```
isPalindrome("madam")
```

Output:

```
true
```

Scenario

Text validation tool.

What is this question asking?

Check if a word reads the same forward and backward.

Concepts Tested

- Functions
 - Strings
 - Logic Building
-

Question 17 — Password Strength Checker

Create:

```
checkPassword(password)
```

Rules:

Password should contain:

- At least 8 characters
- At least 1 number

Return:

```
"Strong"
```

or

```
"Weak"
```

Scenario

Registration page.

Concepts Tested

- Functions
 - Conditions
 - String Analysis
-

Question 18 — Function Returning Function

Create:

```
makeMultiplier(multiplier)
```

Example:

```
let double = makeMultiplier(2);  
  
double(10);
```

Output:

```
20
```

Scenario

Reusable calculation system.

What is this question asking?

A function should return another function.

Concepts Tested

- Higher Order Functions
- Closures (intro)

Question 19 — Callback Function

Create:

```
processUser(name, callback)
```

Example:


```
processUser("Ritik", welcomeUser);
```

Output:

```
Processing User...  
WelcomeRitik
```

Scenario

Backend processing.

What is this question asking?

Pass a function as an argument to another function.

Concepts Tested

- Callbacks
- Higher Order Functions

Question 20 — Shopping Bill Generator

Create:

```
generateBill(items)
```

Example Input:

```
[  
  { name: "Mouse", price: 500 },  
  { name: "Keyboard", price: 1000 },  
  { name: "Monitor", price: 10000 }  
]
```

Output:

Total:11500

Scenario

E-commerce checkout page.

What is this question asking?

Create a reusable function that can calculate the total cost of any shopping cart.

Concepts Tested

- Functions
- Arrays
- Objects
- Real-world Problem Solving

Part 4: Arrays + Objects + Functions (20 Questions)

This section is the most important.

In real-world applications:

- Data is usually stored in **arrays**
- Each item is usually an **object**
- Logic is usually written inside **functions**

So almost every frontend/backend interview problem looks like this:

```
[
  {
    name: "Ritik",
    age: 21
  },

```

```
{  
  name: "Aman",  
  age: 20  
}
```

and you're asked to write functions to manipulate that data.

Easy Level (Q1 - Q7)

Question 1 — Find Adult Users

```
let users= [  
  { name: "Ritik", age: 20 },  
  { name: "Aman", age: 16 },  
  { name: "Priya", age: 25 }  
];
```

Create:

```
getAdults(users)
```

Return only users whose age is 18 or above.

Scenario

A website only allows adults to register.

What is this question asking?

You need to:

1. Loop through users.
2. Check age.

3. Return qualifying users.

Expected Output:

```
[  
  { name: "Ritik", age: 20 },  
  { name: "Priya", age: 25 }  
]
```

Concepts Tested

- Array of Objects
- Functions
- filter()

Question 2 — Product Names Extractor

```
let products = [  
  { name: "Laptop", price: 50000 },  
  { name: "Mouse", price: 500 }  
];
```

Create:

```
getProductNames(products)
```

Return:

```
["Laptop", "Mouse"]
```

Scenario

A dropdown only needs product names.

What is this question asking?

Extract a single property from every object.

Concepts Tested

- Arrays
 - Objects
 - Functions
 - `map()`
-

Question 3 — Find User By Name

```
let users= [  
  { name:"Ritik" },  
  { name:"Aman" },  
  { name:"Priya" }  
];
```

Create:

```
findUser(users,username)
```

Scenario

Search bar functionality.

What is this question asking?

Find a specific object based on a property value.

Concepts Tested

- `find()`

- Functions

Question 4 — Total Marks

```
let students= [  
  { name:"Ritik", marks:80 },  
  { name:"Aman", marks:90 },  
  { name:"Priya", marks:70 }  
];
```

Create:

```
getTotalMarks(students)
```

Return:

```
240
```

Concepts Tested

- reduce()
- Functions

Question 5 — Available Products

```
let products= [  
  { name:"Mouse", stock:10 },  
  { name:"Keyboard", stock:0 },  
  { name:"Monitor", stock:5 }  
];
```

Return only products whose stock is greater than zero.

Scenario

Show only available products.

Concepts Tested

- `filter()`
 - Array of Objects
-

Question 6 — Add New Student

Create:

```
addStudent(students, student)
```

which adds a student to the array.

Scenario

Student admission system.

What is this question asking?

Create a reusable function that updates data.

Concepts Tested

- Arrays
 - Objects
 - `push()`
-

Question 7 — Count Premium Products

```
let products= [  
  { name:"Laptop", price:50000 },
```

```
{ name:"Phone", price:30000 },  
{ name:"Mouse", price:500 }  
];
```

Count products costing more than ₹10,000.

Scenario

Analytics dashboard.

Concepts Tested

- filter()
- length

Moderate Level (Q8 - Q14)

Question 8 — Shopping Cart Total

```
let cart= [  
  { name:"Mouse", price:500, qty:2 },  
  { name:"Keyboard", price:1000, qty:1 },  
  { name:"Monitor", price:10000, qty:1 }  
];
```

Create:

```
getCartTotal(cart)
```

Scenario

Checkout page.

What is this question asking?

For each item:

```
price*qty
```

Add everything.

Expected:

```
12000
```

Concepts Tested

- Arrays
- Objects
- Functions
- `reduce()`

Question 9 — Student Average Generator

```
let students= [  
  { name:"Ritik", marks: [80,90,85] },  
  { name:"Aman", marks: [70,75,80] }  
];
```

Create:

```
getAverageMarks(students)
```

Return average for each student.

Scenario

School Report Card.

Concepts Tested

- Nested Arrays
 - Objects
 - map()
 - reduce()
-

Question 10 — Inventory Search

```
let
inventory= [
  { id:1, name:"Laptop" },
  { id:2, name:"Mouse" },
  { id:3, name:"Monitor" }
];
```

Create:

```
findProductById(id)
```

Scenario

Product detail page.

Concepts Tested

- find()
 - Functions
-

Question 11 — Usernames Generator

```
let users= [
  { name:"Ritik Rajput" },
  { name:"Aman Gupta" }
];
```

Return:

```
[
  "ritik_rajput",
  "aman_gupta"
]
```

Scenario

Registration system.

Concepts Tested

- map()
 - String Methods
 - Functions
-

Question 12 — Highest Scoring Student

```
let students= [
  { name:"Ritik", marks:85 },
  { name:"Aman", marks:95 },
  { name:"Priya", marks:75 }
];
```

Find the highest scorer.

Scenario

School Topper List.

Concepts Tested

- Comparisons
 - Loops
 - Objects
-

Question 13 — Update Product Stock

Create:

```
updateStock(id, quantity)
```

that updates stock of a specific product.

Scenario

Warehouse management.

What is this question asking?

Locate the object.

Update one property.

Concepts Tested

- find()
 - Object Updates
-

Question 14 — Sort Students by Marks

Sort:

```
[
  {name: "A", marks: 70},
  {name: "B", marks: 95},
  {name: "C", marks: 80}
]
```

from highest to lowest.

Scenario

Leaderboard.

Concepts Tested

- `sort()`
 - Array of Objects
-

Hard Level (Q15 - Q20)

These are close to actual frontend interview questions.

Question 15 — Student Grade Report

```
let students= [
  {
    name: "Ritik",
    marks: [80, 90, 85]
  },
  {
    name: "Aman",
    marks: [50, 40, 60]
  }
];
```

Create:

```
generateReport(students)
```

Return:

```
[
  {
    name: "Ritik",
    average: 85,
    grade: "A"
  },
  {
    name: "Aman",
    average: 50,
    grade: "C"
  }
]
```

Scenario

School Report System.

What is this question asking?

For every student:

1. Calculate average.
2. Decide grade.
3. Create a new object.
4. Return a new array.

Concepts Tested

- Arrays

- Nested Arrays
 - Objects
 - Functions
 - map()
 - reduce()
-

Question 16 — Product Revenue Analyzer

```
let products= [  
  {  
    name: "Mouse",  
    price: 500,  
    sold: 20  
  },  
  {  
    name: "Keyboard",  
    price: 1000,  
    sold: 10  
  }  
];
```

Find:

1. Revenue per product
 2. Total Revenue
 3. Best Selling Product
-

Scenario

Business Analytics Dashboard.

Concepts Tested

- Arrays
 - Objects
 - reduce()
 - Comparisons
-

Question 17 — Attendance System

```
let
students= [
  { name:"Ritik", present:true },
  { name:"Aman", present:false },
  { name:"Priya", present:true }
];
```

Create functions:

```
countPresent()
countAbsent()
getPresentStudents()
```

Scenario

School Attendance Portal.

Concepts Tested

- Arrays
 - Objects
 - Functions
 - filter()
-

Question 18 — Library Management System

Store books like:

```
{  
  id:1,  
  title:"Atomic Habits",  
  borrowed:false  
}
```

Create:

```
addBook()  
borrowBook()  
returnBook()  
showAvailableBooks()
```

Scenario

Library Software.

What is this question asking?

Manage a collection of books using functions.

Concepts Tested

- Arrays
- Objects
- Functions
- find()
- filter()

Question 19 — Order Management System

Store orders like:

```
{  
  id:1,  
  customer:"Ritik",  
  amount:5000,  
  status:"Pending"  
}
```

Create:

```
createOrder()  
updateStatus()  
getPendingOrders()  
getCompletedOrders()
```

Scenario

E-commerce Backend.

Concepts Tested

- Arrays
 - Objects
 - Functions
 - Business Logic
-

Question 20 — Mini E-Commerce System (Interview Level)

Create a system with:

```
addProduct()  
removeProduct()  
updateStock()  
purchaseProduct()  
getInventoryValue()
```

Store products like:

```
{  
  id:1,  
  name:"Laptop",  
  price:50000,  
  stock:10  
}
```

Scenario

You are building the backend logic of Amazon/Flipkart inventory.

What is this question asking?

This is not one function.

This is an entire mini project.

You need to:

1. Store products.
2. Add products.
3. Remove products.
4. Update stock.
5. Handle purchases.
6. Calculate inventory value.